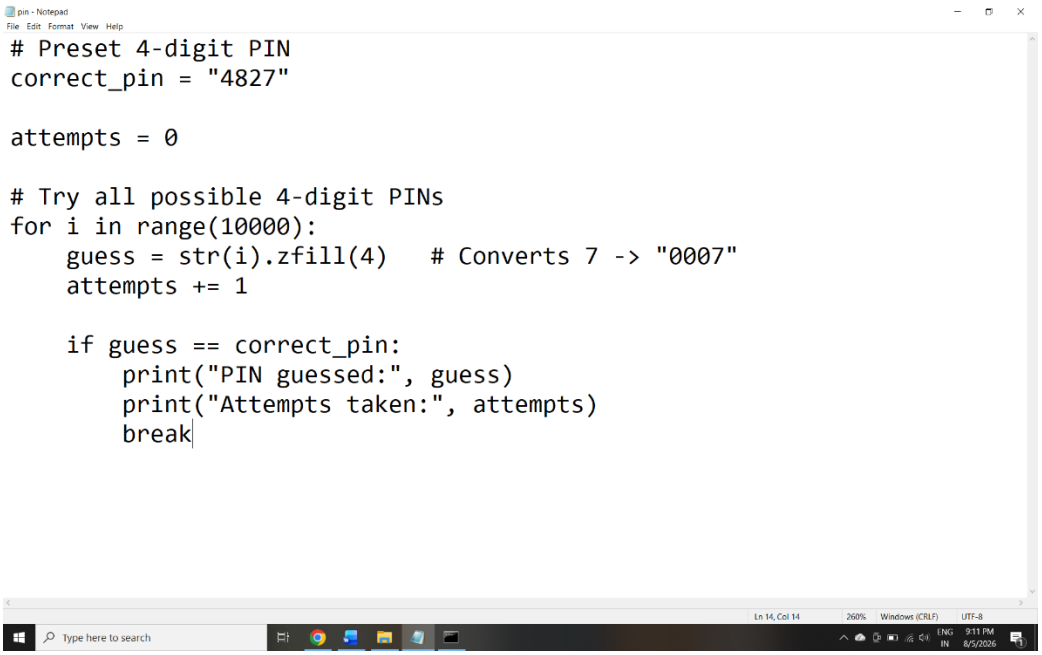
1.Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).

Network Threat	Real-World Example (App/Service)
Phishing	A fake Paytm message asking you to click a link and enter your login details.
DDoS (Distributed Denial of Service)	Instagram becomes temporarily unavailable because attackers flood its servers with traffic.
Man-in-the-Middle (MITM)	Using public Wi-Fi at a café where an attacker intercepts your WhatsApp or browser traffic.
Malware	Downloading a fake APK that steals WhatsApp or Paytm account information.
Password Attack	An attacker tries many password combinations to log into your Instagram account.

2.Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value.Constraint: The script should print the number of attempts it took to guess the correct PIN.



pin - Notepad

File Edit Format View Help

```
# Preset 4-digit PIN
correct_pin = "4827"

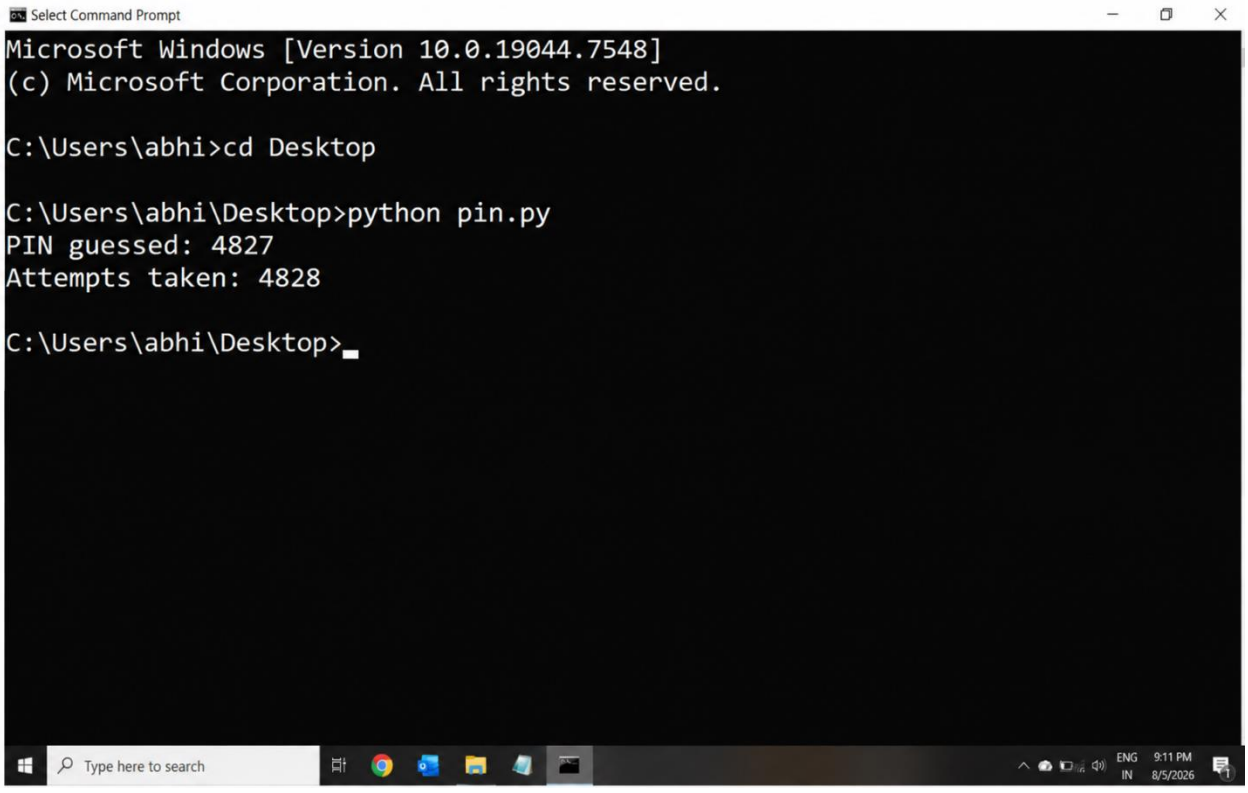
attempts = 0

# Try all possible 4-digit PINs
for i in range(10000):
    guess = str(i).zfill(4)   # Converts 7 -> "0007"
    attempts += 1

    if guess == correct_pin:
        print("PIN guessed:", guess)
        print("Attempts taken:", attempts)
        break
```

Ln 14, Col 14 260% Windows (CRLF) UTF-8

Type here to search ENG IN 9:11 PM 8/5/2026



Select Command Prompt

```
Microsoft Windows [Version 10.0.19044.7548]
(c) Microsoft Corporation. All rights reserved.

C:\Users\abhi>cd Desktop

C:\Users\abhi\Desktop>python pin.py
PIN guessed: 4827
Attempts taken: 4828

C:\Users\abhi\Desktop>
```

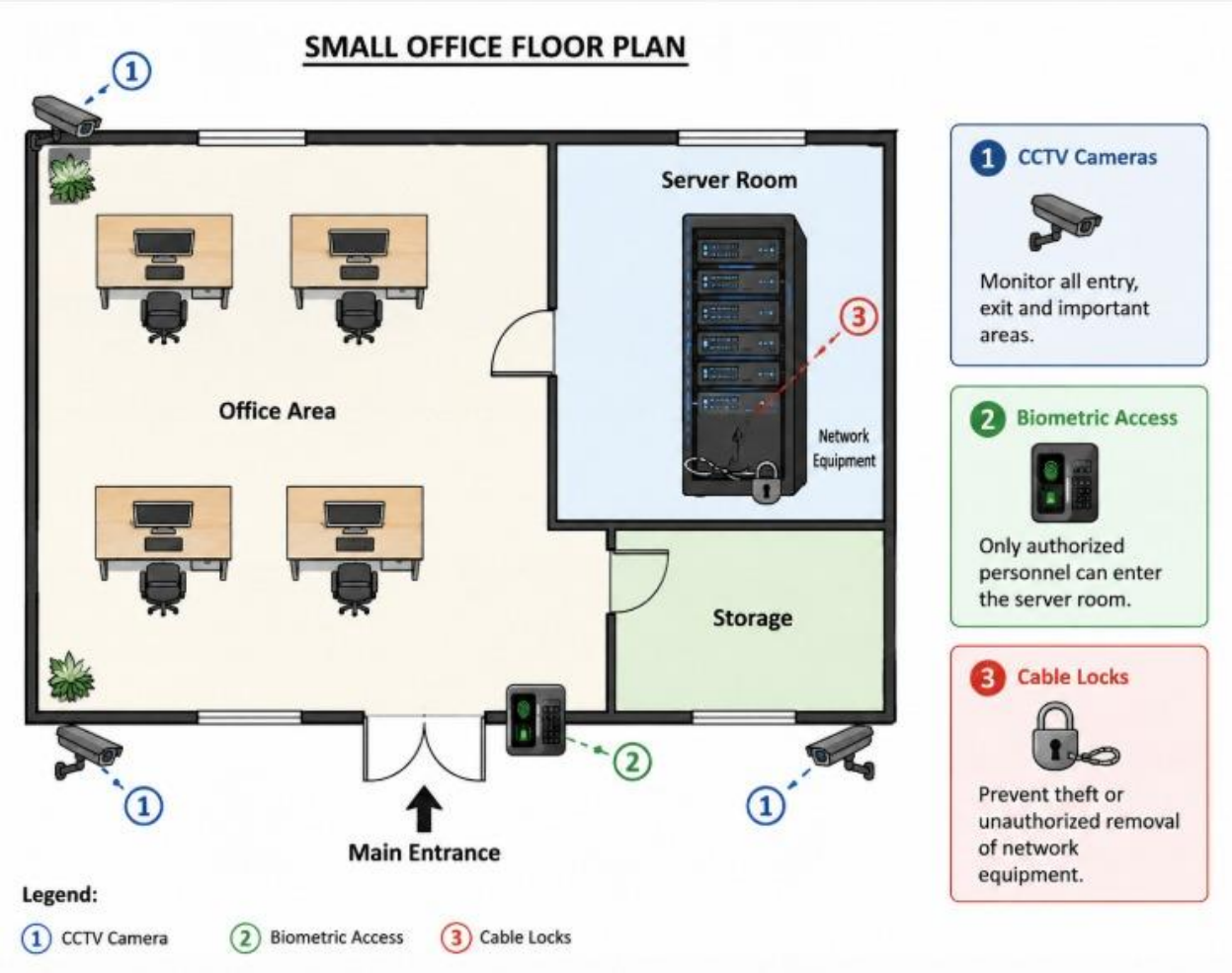
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3.List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.

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Physical Security Risk	Mitigation Step
Unauthorized access to the server room	Use biometric or key-card access control for entry.
Fire outbreak	Install fire detection and automatic fire suppression systems.
Power failure	Use UPS (Uninterruptible Power Supply) and backup generators.
Theft of servers or equipment	Install CCTV cameras and keep servers in locked racks.

4.Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).



5.Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.

College Computer Lab Network & Physical Security Checklist

Action	Why it helps
Keep antivirus software updated on all lab computers	Prevents malware and virus infections from USB drives and downloaded files.
Use strong passwords for student, faculty, and admin accounts	Reduces the risk of unauthorized logins and password attacks.
Enable a firewall on every computer and the lab router	Blocks suspicious network traffic and external attacks.
Restrict access to the server room and network switch cabinet	Protects critical networking equipment from tampering or theft.
Install CCTV cameras covering entrances and network equipment areas	Helps monitor unauthorized access and supports incident investigation.
Lock network racks and secure cables with cable locks	Prevents equipment theft and accidental cable disconnection by students.
Regularly back up lab systems and important academic data	Ensures quick recovery after ransomware, hardware failure, or accidental deletion.